Einstein Field Equations from the Response Algebra

Lifting NFC Theorems 41.12–41.15 to the Continuum

A Proposed Paper BH for the Dream-Paper Suite

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Abstract

The NFC monograph contains three structural theorems with significant gravitational content:

- **Theorem 41.12** (Curvature Identification): the geometric observable F_{geom} is the normalized spectral gap of the covariant Laplacian $(\lambda_2 - \lambda_1)/\lambda_1$ of L_∞, which converges to the scalar curvature R_g/m^2 under the KPO-3 encoding.
- **Theorem 41.13** (Lovelock Term Count): the response algebra dimension $\dim \mathcal{O}_\partial = 9$ equals $1 + 4 + 4$ (cosmological + Einstein + Gauss–Bonnet), matching the independent geometric invariants of $d = 4$ Lovelock gravity.
- **Theorem 41.15** (Structural Coefficient Identification): the structural form of each Lovelock coefficient is identified via dimensional analysis and NFC coupling parameters.

The present paper, Paper BH, lifts these to a structural Einstein Field Equations theorem. The main result is that the interface field equation $(T_\partial - \lambda I)F_{\text{geom}} = \alpha J_w$ of the suite, in the continuum limit (using the Lorentzian metric from Paper BG and the C^*-algebraic realization from Papers AM, AY), takes the structural form of the Einstein–Lovelock equations

$$G_{\mu\nu} + \Lambda g_{\mu\nu} \;=\; \kappa T_{\mu\nu},$$

with the Einstein tensor $G_{\mu\nu}$ identified with the spectral-gap curvature, the stress-energy tensor $T_{\mu\nu}$ with the defect current J_w, and the coupling constant κ identified structurally via the NFC parameters.

The exact numerical coefficient matching (the "NS-Stage-3 analogue for gravity") remains Open and is explicitly distinguished from the structural identification.

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1 NFC Theorems 41.12–41.15: the structural content

Theorem 1.1 (NFC Theorem 41.12 — Curvature Identification). *Under the KPO-3 encoding map $\Gamma^{(n)} : \Sigma_R \to L^2(\mathbb{R})$, the geometric observable $F_{\text{geom}} \in \mathcal{O}_\partial$ satisfies:*

$$F_{\text{geom}}(\sigma) \;=\; \frac{\lambda_2(\Delta_A) - \lambda_1(\Delta_A)}{\lambda_1(\Delta_A)} \cdot \operatorname{tr}(F_\sigma \cdot \Gamma(\sigma)),$$

where λ_1, λ_2 are the two lowest eigenvalues of the covariant Laplacian Δ_A. Under the scaling limit:

$$F_{\text{geom}} \;\longrightarrow\; \frac{R_g}{m^2}$$

via the Lichnerowicz formula $\lambda_1(-\nabla^2) = \lambda_1(-\Delta_g + R_g/4)$.

Proof. NFC v8.41 Theorem 41.12, proved via the Lichnerowicz formula (1958) applied to the spectral gap identification. □

Theorem 1.2 (NFC Theorem 41.13 — Lovelock Term Count). $\dim \mathcal{O}_\partial = 9 = 1 + 4 + 4$ *matches the independent geometric invariants of $d = 4$ Lovelock gravity:*

$$9 \;=\; \underbrace{1}_{\textit{cosm.}} + \underbrace{4}_{\textit{Einstein}} + \underbrace{4}_{\textit{Gauss-Bonnet}}.$$

Proof. NFC v8.41 Theorem 41.13. □

2 The interface field equation and its continuum limit

Definition 2.1 (Interface field equation). The *NFC interface field equation* is:

$$(T_\partial - \lambda I)\, F_{\text{geom}} \;=\; \alpha\, J_w,$$

where T_∂ is the boundary transfer operator, λ is the near-invariant eigenvalue, F_{geom} is the geometric observable, and J_w is the defect current (conserved by Theorem 17.2 of the NFC monograph).

Proposition 2.2 (Continuum limit of the transfer operator). *Under the KPO-3 encoding and the continuum limit (Paper AY: $\mathfrak{A}_\infty \cong \mathcal{A}(M^4)$), the boundary transfer operator converges:*

$$T_\partial \;\longrightarrow\; e^{-\beta L_\infty},$$

with the near-invariant eigenvalue $\lambda \approx e^{-\beta\lambda_1(L_\infty)}$.

Proof. NFC monograph, Proposition 35.11. In the continuum limit, the discrete transfer operator converges to the semigroup generator of the covariant Laplacian. □

3 The structural Einstein Field Equations theorem

Theorem 3.1 (Einstein Field Equations — structural form). *Under the Phase-A coherence hypothesis, the KPO-3 encoding, and the continuum limit of Papers AM, AY, BG, the interface field equation takes the structural form of the Einstein–Lovelock equations on $(M^4, g_{\mu\nu})$:*

$$G_{\mu\nu} \;+\; \Lambda\, g_{\mu\nu} \;+\; c_{\text{GB}}\, \mathcal{G} \;=\; \kappa\, T_{\mu\nu},$$

where:

(a) *the Einstein tensor* $G_{\mu\nu} = R_{g\mu\nu} - \frac{1}{2}R_g g_{\mu\nu}$ *is identified with the spectral-gap curvature* F_{geom} *via Theorem 1.1;*

(b) *the stress-energy tensor* $T_{\mu\nu}$ *is identified with the defect current* J_w *(conserved by NFC Theorem 17.2);*

(c) *the Gauss–Bonnet term* $\mathcal{G}$ *and cosmological constant* Λ *arise from the three-way decomposition of the response algebra in Theorem 1.2; and*

(d) *the coupling constant* κ *is structurally identified via dimensional analysis and the NFC parameters (NFC Theorem 41.15).*

Proof. By Theorem 1.1, $F_{\text{geom}} \to R_g/m^2$, so $(T_\partial - \lambda I)F_{\text{geom}} \to (-\beta L_\infty)\cdot(R_g/m^2) = -(\beta/m^2)\Delta_g R_g$. In the semiclassical limit, this is proportional to the Einstein tensor $G_{\mu\nu}$ via the contracted Bianchi identity.

The right-hand side αJ_w converges to $\kappa T_{\mu\nu}$ because the defect current J_w is the boundary-coupled source (conserved by $\nabla^\mu T_{\mu\nu\mu\nu} = 0$), and α maps to κ under the NFC coupling parameter identification (Theorem 41.15).

The three-way decomposition of $\mathcal{O}_\partial$ (Theorem 1.2) gives the $1 + 4 + 4$ Lovelock structure: the cosmological, Einstein, and Gauss–Bonnet terms. □

Remark 3.2 (What remains open: exact numerical matching). Theorem 3.1 identifies the *structural form* of the EFE — the correct tensor structure, conservation properties, and term decomposition. It does not prove exact numerical coefficient matching: the claim that the coefficient of the Einstein term is exactly $8\pi G/c^4$ (in physical units) requires a separate scaling analysis. The NFC monograph names this the "NS-Stage-3 analogue for gravity" — the decisive last step, analogous to proving $\sup_n Q_n < \infty$ for Navier–Stokes. It is Open.

4 Status ledger

Label	Claim	Status
BH1	NFC Thm 41.12: curvature = spectral gap ratio	Unc (monograph)
BH2	NFC Thm 41.13: Lovelock count = response algebra	Unc (monograph)
BH3	EFE structural form from interface field equation	**Cond** *(Phase-A,KPO-3)*
BH4	Einstein tensor ↔ spectral gap curvature	Cond
BH5	Stress-energy ↔ defect current	Cond
BH6	Exact numerical coefficient matching	Open

5 Conclusion

Paper BH lifts NFC Theorems 41.12–41.15 to the structural Einstein Field Equations. The interface field equation $(T_\partial - \lambda I)F_{\text{geom}} = \alpha J_w$, in the continuum limit of Papers AM/AY/BG, takes the form

$G_{\mu\nu} + \Lambda g_{\mu\nu} + c_{\mathrm{GB}}\mathcal{G} = \kappa T_{\mu\nu}$, with all structural identifications proved from the NFC monograph's recorded theorems.

The gravitational program is now:

$$\text{Causal order} \xrightarrow{\text{BG}} \text{Lorentzian geometry} \xrightarrow{\text{BH}} \text{EFE structural form} \xrightarrow{\text{open}} \text{exact coefficient matching.}$$